Skanda Shreesha Prasad

M.S. Computer Science Student

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Education

M.S. **New York University - Courant Institute of Mathematical Sciences**, Computer Science — New York, NY, July 2025 – present

- Coursework: Deep Learning, Computer Vision, Bayesian Machine Learning, Fundamental Algorithms, Operating Systems

B.Tech.PES University, Computer Science and Engineering — Bengaluru, India, Sept 2021 – May 2025

Research Experience

New York University, Hierarchical Planning, H-JEPA and World Models — New York, NY, Mar 2026 – present
Prof. Yann LeCun and Prof. Jean Ponce

- Researching hierarchical JEPA world models for hierarchical planning and reasoning.
- Developing dynamic temporal abstraction methods that learn when low-level transitions should be grouped into higher-level states and actions instead of relying on fixed-stride hierarchies.
- Evaluating hierarchical latent planning on robotic manipulation environments.

New York University, Multiscale JEPA for Self-Supervised Active Matter Representation Learning — New York, NY, Apr 2026 – Apr 2026
Prof. Yann LeCun and Prof. Alfredo Canziani

- Built global, spatial, and multiscale JEPA architectures for self-supervised representation learning on active-matter simulations under a 100M-parameter constraint.
- Evaluated frozen representations using linear and KNN probes; placed second on the course leaderboard and presented the project as an Outstanding Project.

New York University, Multimodal Protein Retrieval from Sequences and Cryo-EM Density Maps — New York, NY, Feb 2026 – present
Prof. David Fouhey

- Developing a multimodal representation framework that jointly embeds amino-acid sequences and 3D cryo-EM density maps into a shared latent space.
- Using a JEPA-style cross-modal predictive objective for sequence-structure retrieval with weakly aligned biological data.

New York University, Active Spectral Learning for Non-Stationary Gaussian Processes — New York, NY, Sept 2025 – Dec 2025
Prof. Andrew Gordon Wilson

- Built an active-learning pipeline for Gaussian Processes with Generalized Spectral Mixture kernels, using ensemble disagreement to guide sampling in non-stationary environments.
- Compared Monte Carlo epistemic-variance acquisition with BALD and other baselines, analyzing structural uncertainty, sampling bias, and predictive performance.

PES University, Code-Mixed LLM for Natural Language Generation and Translation — Bengaluru, India
Prof. Dr. Uma D — Oct 2023 – Jan 2025

- Built an English-to-Hindi-English code-mixed translation system and developed English-Kannada code-mixing data for low-resource language modeling in a university-funded project.

Experience

The Bosch Group, AI Research Intern — Bengaluru, India
Mar 2025 – July 2025
5 months

- Researched weather-resilient ADAS perception for rain, fog, and low-visibility conditions; work resulted in three Indian patent applications.

The Bosch Group, Generative AI Intern — Bengaluru, India
June 2024 – Aug 2024
3 months

- Built a generative-AI-powered chatbot for an upcoming BGSW Data Accelerator system, including metadata management and information retrieval.

The Bosch Group, Machine Learning and Computer Vision Intern — Bengaluru, India
June 2023 – July 2023
2 months

- Developed computer-vision models for diesel-engine production-line automation and microscopic-part classification.

PES University - Department of Computer Science and Engineering, Teaching Assistant — Bengaluru, India
July 2024 – May 2025
11 months

- Teaching Assistant for Generative AI and its Applications and Mathematics for Computer Science Engineers.
- Prepared slide decks and coding assignments, guided course projects, and evaluated assignments.

Publications and Patents

Enhancing Multi-Agent LLM Frameworks: A Context-Aware Approach for Securing Dynamic Agent Collaboration — Jan 2025
Conference publication at ICAART 2025.
Skanda Shreesha Prasad
(ICAART)

Indian Patent Applications 202541105316, 202541093849, and 202641023935 — Jan 2025
Patent applications from research on weather-resilient ADAS perception.
Bosch Research
(The Bosch Group)

Skills

Programming: Python, C, Java, SQL, MATLAB, R

ML / Research: PyTorch, JAX, Computer Vision, Representation Learning, World Models, JEPA, LLMs, Gaussian Processes

Tools: AWS, Docker, Kubernetes, Slurm